**MATH 1342 – Elementary Statistics – Week 7: Confidence Intervals and the Bootstrap**

**Oct 6 – 8** | 3.2 Understanding and Interpreting Confidence Intervals • 3.3 Constructing Bootstrap Confidence Intervals Using Standard Error | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

| Module Objectives | Instructional Materials | Learning Activities | Assessments | Tools |
|---|---|---|---|---|
| **1.** Construct a 95% confidence interval for a mean or proportion as statistic ± 2·SE, given the statistic and its standard error. (CLO 6)<br>**2.** Interpret a confidence interval in context, explain the confidence level as the long-run capture rate of the method, and identify common misinterpretations. (CLO 4, CLO 6) | e-Text: 3.2 Understanding and Interpreting Confidence Intervals<br>Interactive Lecture: 3.2 Understanding and Interpreting Confidence Intervals<br>Slides: 3.2 Understanding and Interpreting Confidence Intervals | Participation: Interpreting Confidence Intervals – which of 20 simulated CIs capture μ; fix three misinterpretations | Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM) | WileyPLUS<br>StatKey<br>Blackboard<br>Zoom (live class) |
| **3.** Describe how a bootstrap sample and a bootstrap distribution are created, and explain why the bootstrap distribution is centered at the sample statistic. (CLO 5, CLO 6)<br>**4.** Use StatKey to generate a bootstrap distribution and estimate the standard error as its standard deviation. (CLO 5, CLO 6)<br>**5.** Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CLO 6) | e-Text: 3.3 Constructing Bootstrap Confidence Intervals<br>Interactive Lecture: 3.3 Constructing Bootstrap Confidence Intervals<br>Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator)<br>StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals | Participation: Bootstrap a Proportion – Try It: 32 of 50 prefer Brand A; build the bootstrap distribution, read off SE, form the CI | Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM) | WileyPLUS<br>StatKey<br>Blackboard<br>Zoom (live class) |
| *Course resources and assessments of earlier weeks (not aligned to this week's objectives)* | | | Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts) | Assessment Tool in Blackboard<br>Respondus LockDown Browser and Monitor |

**CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals.